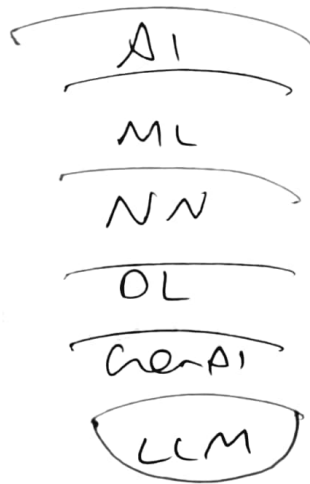


Subset of AI,



AI :-

- ↳ Machine that Simulator human intelligence
- Analyse Patterns and data to make decision

ML :-

- ↳ System that learn from data without explicit programming
- Discover hidden Pattern and make Prediction

Neural Network

- ↳ ML model that mimic that complex function of the human
- , These model consist of interconnect nodes or neurons that process data.

DL:

→ Adversarial learns with multi-layered neural network like the human brain

Generative AI:

→ Gen AI learns patterns from existing data and generates new content

LLM:

→ An LLM is a transform-based AI model that learns language patterns from large text datasets and generates human-like responses.

LLM in detail:

A large neural network trained on huge amount of text to understand and generate human-like language.

Why LLM?

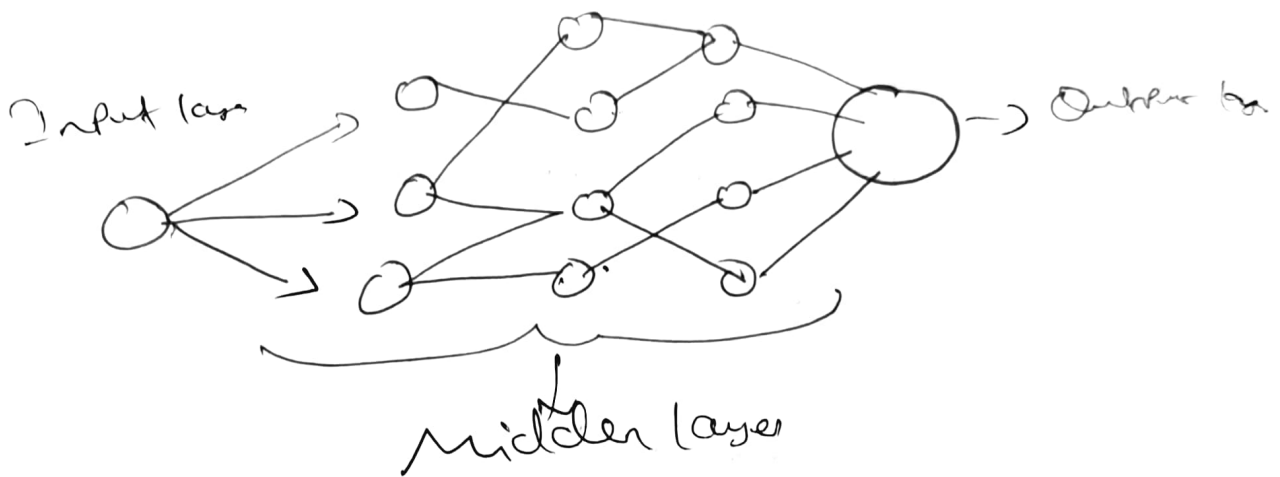
Before LLM NLP Struggled

Ex: Bank $\angle$ ^{financial} _{insurance}

Neural Networks (inspired by human brain)

-) It is the fundamental building block of ~~all~~ deep learning

Structure



Input layer: -) Receives the feature
Hidden layer -> where learning happens

$$\text{Output} = \text{Activation} \left(\sum (\text{Input} \times \text{weight}) + \text{Bias} \right)$$

Output layer -> produces the prediction

Key term!

weights $\Rightarrow$ It determines how important each input is

Bias $\Rightarrow$ shifts the output

Activation function!

It decides whether the neuron should be activated! and how stronges it should pass information to the next layer!

1) ReLU ~~for~~

$$\text{ReLU}(x) = \max(0, x)$$

$$\text{Range } (0 - \infty)$$

2) Sigmoid

$$f(x) = \frac{1}{1 + e^{-x}}$$

$$\text{for } (0 \text{ to } 1)$$

3) Tanh:

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad \text{range } (-1 \text{ to } 1)$$

4) Softmax

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}} \quad \begin{matrix} 0 \text{ to } 1 \\ \text{Sum} = 1 \end{matrix}$$

forward Pass:

- It is the Process of Passing input data
Through a neural network to generate
Predictions.

Loss fun:

Measure how wrong the model's prediction
is (MSE, Cross Entropy loss)

Back Propagation

It is a rule that calculates how
much weight contributed to the error

Gradient Descent $\rightarrow$ It is an optim algorithm that updates
the weights to reduce the loss

LSTM:

It is a type of Recurrent Neural Network
designed to remember information for a long time
Solve for:

Combrant:

Memorable Gate:

Cell State acts like long term mem

Forget Gate:

Decide what info to remove from mem

Input Gate:

Decide what new information should be stored

Output Gate

Decide what should be passed to next step

Limitation LSTM

() High computation cost

→ Limited long term mem

→ Overfitting!

CRV:

→ simplified version of LSTM

Update Gate:

Decides how much of the previous info should be kept and how much new info

Reset Gate

Decides how much past info should be ignored

limiter:

Sequential Process (slow)

Transformer:

"Attention All you need" - 2017

⇒ It is a DL architecture that uses self-attention mechanism to process all words in parallel and understand the relationships between them

why!?

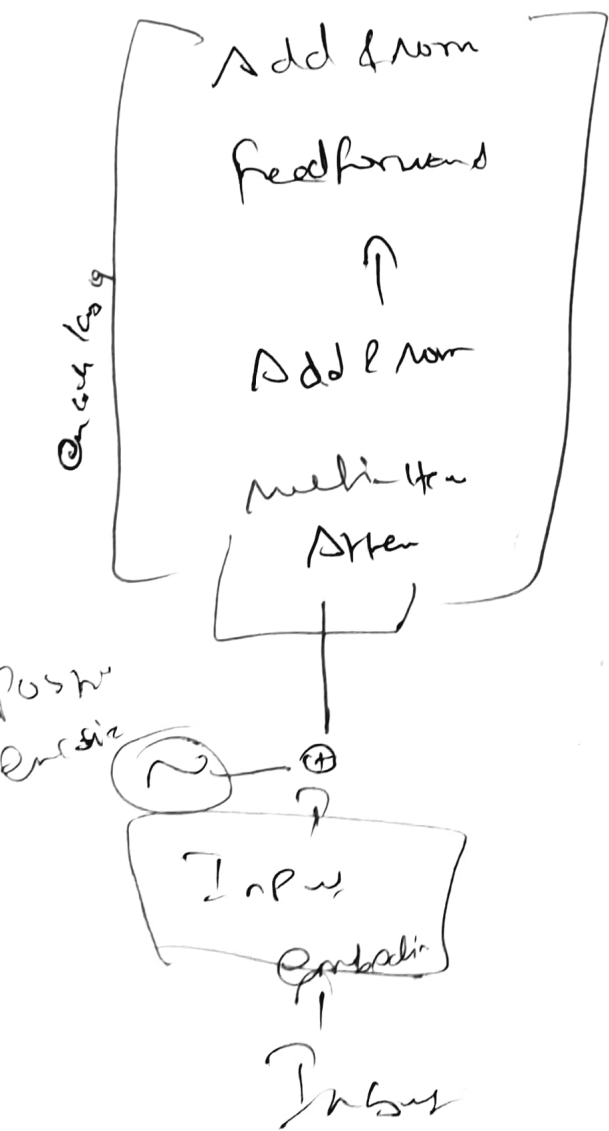
↳ Instead of memorizing word sequences,

allow every word to look at every other word using attention

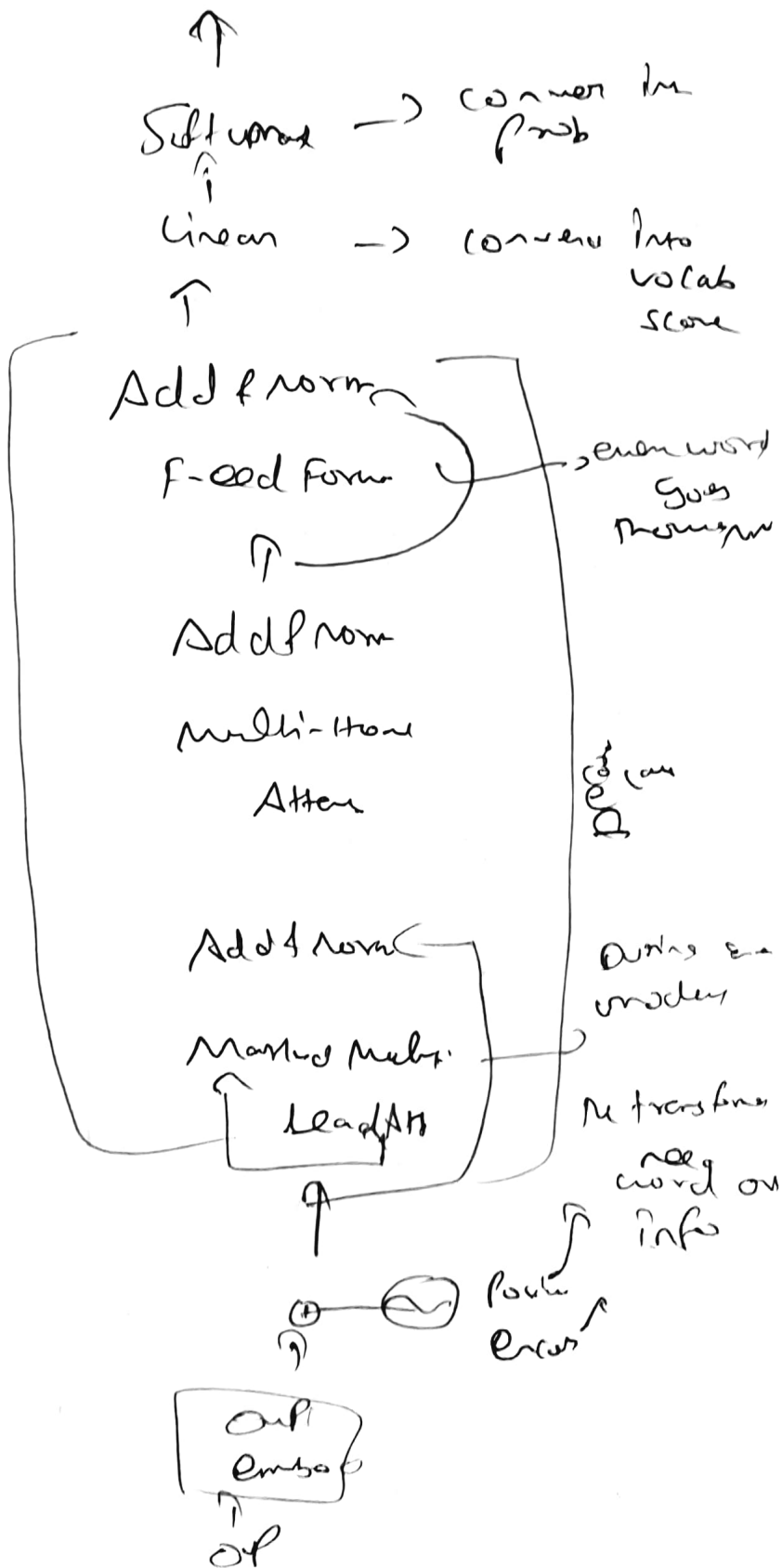
Transformer Architecture

This contains residual connect

Understand relationship between words



Output Prob



Attention uses three vectors

Query $\rightarrow$ what am I looking for

Key $\rightarrow$ what info do I contain

Value $\rightarrow$ what info do I provide

$$\text{Attention} = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Self Attention:

because only transforms

(Input $\rightarrow$ Tokens $\rightarrow$ Token embeddings $\rightarrow$ PE $\rightarrow$)

Transformer Decoder block $\rightarrow$ linear $\rightarrow$ softmax
Next token pred]

Limitations:

$\hookrightarrow$ Hallucination

$\hookrightarrow$ Knowledge limitation

BERT:

Designed mainly for understanding tasks

BERT: training on:

$\hookrightarrow$ Masked Language Model

$\hookrightarrow$ Next Sentence Prediction

Limiter...

- ↳ Expose fire tuning
- ↳ High Computation cost

Parameters $\rightarrow$ Internal values that a Neural Network (learns)

Tokens $\rightarrow$ Smaller word possible (in process)

State of LLM:

Pretraining $\rightarrow$ fine $\rightarrow$ tuning $\rightarrow$ Alignment

Key concept

Temperature

Top K - Sampling

Top P - Sampling

(Hallucination)

Limit of LLM

↳ Hallucination

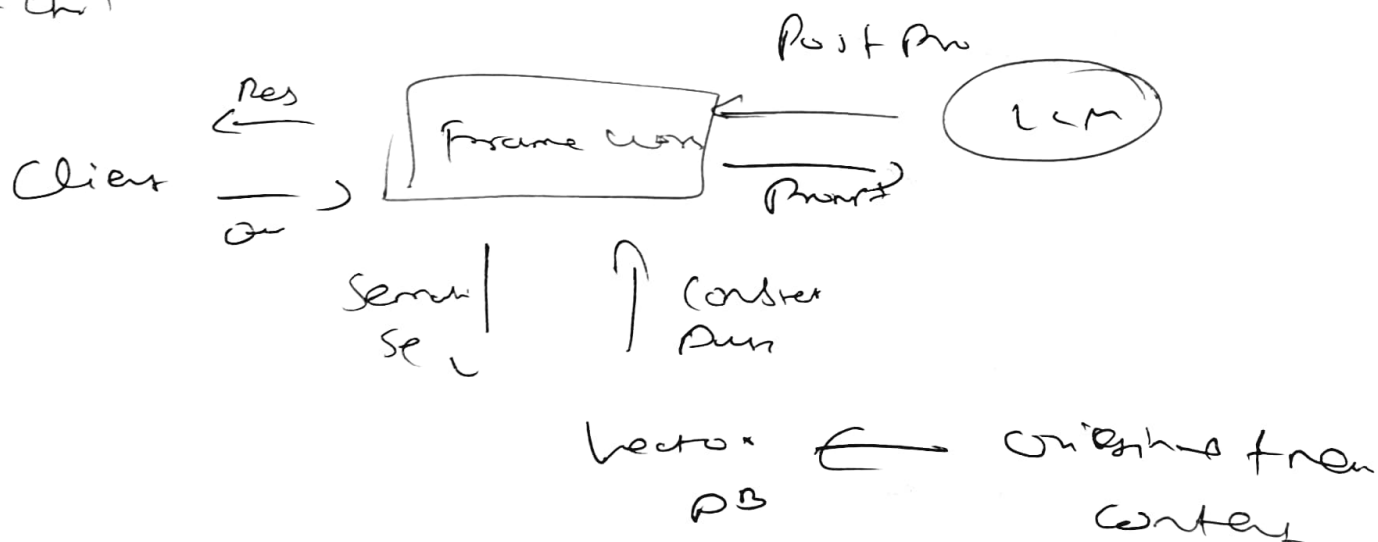
Missclassification

RAG!

It is a tech that improves LLM by giving them access to external knowledge so

$$RAG = LLM + search + engine + DB$$

Arch:



Process

(input data) → (char) → embed → vector

Types of Retrieval

Deny - span (used as user)

was embed

Agentic AI:

System that can independently pursue goals
by planning meta-levels - using tools, a

(as:

↳ Goal-oriented

↳ Planning

↳ Reasoning

→ Tool use

→ memory

→ adaptation

Standard work flow

Goal → Understand → Plan → Execute -

observe → Evaluate → Improve →
(complete)

State: → current progress

Creating an AI Agent

- 1) Define goal
- 2) Choose a LLM
- 3) Create the state
- 4) Create Tools
- 5) Create Agent workflow
- 6) Add Memory
- 7) Add Decision Making
- 8) Implement using frameworks

Multi-agent AI system

A complex task divided among multiple agents

Super Agent $\rightarrow$ Main controlling agent

Sub Agent $\rightarrow$ designed to perform a specific task

Ex: draft an email

Goal $\rightarrow$ write an email

Planner $\rightarrow$ create draft, steps

Super Agent $\rightarrow$ coordinates tasks

Sub Agent $\rightarrow$ Research, write

Tools $\rightarrow$ Grammar checker, calendar, Email API
State: Store draft, progress
Feedback loop $\rightarrow$ improve email quality